



NUTRITIONAL DISORDERS & METABOLIC SYNDROME

COPYRIGHT © 2025 NURSEACHIEVE, CORP.

1

TABLE OF CONTENTS

- [Objectives](#)
- [Definitions](#)
- [Obesity](#)
 - [Pathophysiology](#)
 - [Etiology](#)
 - [Diagnosis](#)
 - [Manifestations](#)
 - [Complications](#)
 - [Goals of Care](#)
 - [Interventions](#)
- [Metabolic Syndrome](#)
 - [Definition](#)
 - [Diagnosis](#)
 - [Pathophysiology](#)
 - [Etiology](#)
 - [Management](#)
- [Adult Malnutrition](#)
 - [Definition](#)
 - [Etiology](#)
 - [Pathophysiology](#)
- [Manifestations](#)
- [Goals of care](#)
- [Interventions](#)
- [Complications](#)
- [Summary](#)
- [References](#)

COPYRIGHT © 2025 NURSEACHIEVE, CORP.

2

OBJECTIVES

- To understand the indications of body mass index measurements in the adult population
- To outline the pathophysiology related to obesity and its multi-factorial etiology
- To recognize the effect of obesity on multiple body systems, and the management guidelines to reduce such effects
- To recognize specific diagnostic criteria for metabolic syndrome and its effect on health outcomes
- To define adult malnutrition and outline relevant etiologies
- To understand the pathophysiology of starvation and its manifestations
- To implement treatment for adult malnutrition and to monitor for associated complications

COPYRIGHT © 2025 NURSEACHIEVE, CORP.

3

DEFINITIONS

Body Mass Index (BMI): a measure of height for weight (kg/m^2), determined by dividing body weight (kg) by the square of the height (m).



Underweight:
 $<18.5 \text{ kg}/\text{m}^2$

Normal weight:
 $18.5\text{-}24.9 \text{ kg}/\text{m}^2$

Overweight:
 $25\text{-}29.9 \text{ kg}/\text{m}^2$

Obese:
 $>30 \text{ kg}/\text{m}^2$



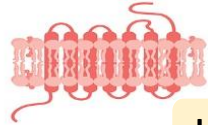
COPYRIGHT © 2025 NURSEACHIEVE, CORP.

4

OBESITY - PATHOPHYSIOLOGY

- **Neuropeptide Y:** stimulates appetite
- **Ghrelin:** stimulates appetite
- **Leptin:** suppresses appetite

Hormonal appetite regulation imbalance



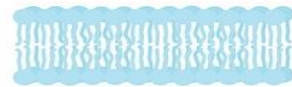
Large volume of lipids to be stored accumulates

Primary obesity:
excess calorie intake compared to body energy expenditure

Secondary obesity:
results from congenital anomalies

Increase in the size and number of adipocytes, located primarily in visceral and subcutaneous tissues. Preadipocytes become adipocytes

Adipocyte hypertrophy and hyperplasia



COPYRIGHT © 2025 NURSEACHIEVE, CORP.

5

OBESITY- ETIOLOGY

PSYCHOSOCIAL FACTORS

Social factors: culture, ethnicity

Associations other than hunger prompting food consumption (reward, comfort etc.)

Increased accessibility of food, processing and packaging

ENVIRONMENTAL FACTORS

Lack of physical exercise (technology, transportation)

Low socioeconomic status



MULTI-FACTORIAL ETIOLOGY OF OBESITY

Studies suggest genetic links in the development of obesity, more research is required in this area

GENETIC FACTORS

COPYRIGHT © 2025 NURSEACHIEVE, CORP.

6

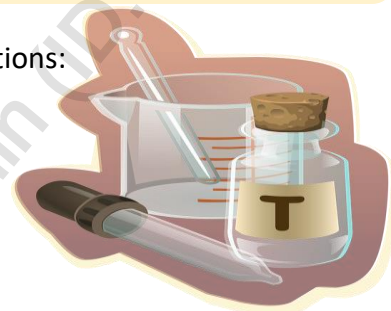
OBESITY- DIAGNOSTICS

BMI is the most common measure of obesity. However, this value must be considered in relation to the client's age, gender, and body composition.

- **Waist circumference:** Increased risk for men >30 inches, women >35 inches
- **Body shape:** Apple shaped bodies (fat in abdominal area) have a greater complication risk than pear shaped bodies (fat in upper thigh area)
- **Hydrostatic weighing and bioelectrical impedance** determines body fat percentage

Additional tests for diagnosis and assessing associated complications:

- Thyroid profile
- Serum glucose: determination of co-existing diabetes
- Serum cholesterol, lipid profile (depressed HDL, elevated LDL)
- Electrocardiogram: detects effect of obesity on the heart



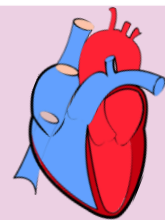
COPYRIGHT © 2025 NURSEACHIEVE, CORP.

7

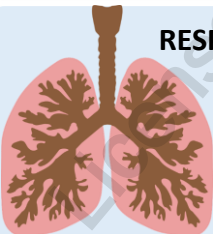
MANIFESTATIONS AND COMPLICATIONS

CARDIOVASCULAR: Obese clients are at a high risk for cardiovascular disease

- Hypertension
- Hyperlipidemia
- Stroke
- Peripheral artery disease
- Left ventricular hypertrophy
- Atrial fibrillation
- **Metabolic syndrome***

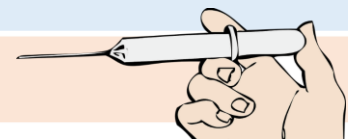


RESPIRATORY: inflammation and fat around the diaphragm creates complications



- Obesity hypoventilation syndrome
- Sleep apnea
- Asthma
- Pulmonary hypertension
- Exercise intolerance

ENDOCRINE: increased risk for type 2 diabetes mellitus due to the development of insulin resistance



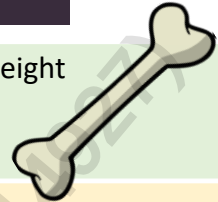
COPYRIGHT © 2025 NURSEACHIEVE, CORP.

8

MANIFESTATIONS and COMPLICATIONS

MUSCULOSKELETAL: complications arise from increased inflammation and weight

- Osteoarthritis
- Impaired mobility
- Gout
- Chronic pain



GASTROINTESTINAL: high cholesterol and increased abdominal weight

- Gallstones
- GERD
- Colon cancer

REPRODUCTIVE

Female: menstrual irregularities, infertility, gestational diabetes

Male: hypogonadism, sexual dysfunction



PSYCHOSOCIAL: stigmatization and discrimination of obese people is common

- Depression
- Low self esteem
- Social isolation
- Withdrawal

COPYRIGHT © 2025 NURSEACHIEVE, CORP.

9

OBESITY - GOALS OF CARE

1

Modify nutritional patterns

2

Modify exercise patterns

3

Achieve and maintain prescribed weight loss

4

Prevent and treat obesity-related complications



COPYRIGHT © 2025 NURSEACHIEVE, CORP.

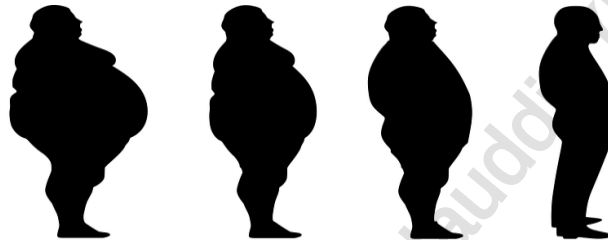
10

OBESITY - NON-PHARMACOLOGICAL INTERVENTIONS

Exercise Planning: physical activity decreases appetite, promotes self-esteem, and increases the basal metabolic rate

Guidelines for physical activity for adults

- Aerobic Physical Activity: 150 minutes per week of moderate intensity, or 75 minutes per week of vigorous intensity, or an equivalent combination
- Muscle-strengthening activities: 2 or more days per week, at least one set of 8-12 repetitions for all major muscle groups



COPYRIGHT © 2025 NURSEACHIEVE, CORP.

11

OBESITY - NON-PHARMACOLOGICAL INTERVENTIONS

ACTIVITY EXAMPLES

Moderate Intensity

- Brisk walking
- Water aerobics
- Ballroom dancing
- General gardening



Vigorous Intensity

- Jogging/running
- Swimming laps
- Jumping rope
- Fast bicycling



Muscle-Strengthening

- Weight lifting
- Resistance bands
- Body weight exercises
- Yoga



COPYRIGHT © 2025 NURSEACHIEVE, CORP.

12

OBESITY - NONPHARMACOLOGICAL INTERVENTIONS

Nutrition: dietary plans will vary greatly from client to client and are based on specific needs. General principles of nutritional therapy for clients with obesity include:

- **3500 calories is typically equivalent to one pound of weight lost**
- Dieting, such as fad diets, are usually unbalanced and are not effective or sustainable means of losing weight. Use of these diets will result in deficiency of one or more nutrients
- Regular meals should be eaten in small servings, typically should contain lower amounts of fat and calories, and contain adequate amounts of vitamins, minerals, and fiber
- Realistic and healthy weight loss goals (such as 1-2 pounds per week) are important to discuss with the client



COPYRIGHT © 2025 NURSEACHIEVE, CORP.

13

OBESITY - NON-PHARMACOLOGICAL INTERVENTIONS

Research has shown that overweight people are stimulated by external factors to consume food. **Behavior modification** focuses on examining the lifestyle of clients and creates strategies to control food cues. Examples may include:

- Keeping food records
- Avoiding “mindless” eating, such as while watching television or reading
- Eating slowly and drinking water with your meals
- Using non-food rewards for meeting goals



Support groups: are often helpful, as weight loss success can be promoted through peer support. Weight loss programs are often expensive and may not be accessible for a low-income client.

COPYRIGHT © 2025 NURSEACHIEVE, CORP.

14

OBESITY - PHARMACOLOGICAL INTERVENTIONS

Pharmacotherapy for weight reduction is indicated in clients with a BMI > 30 kg/m², or a BMI of 27 kg/m² with comorbidities.

Pharmacologic therapy may involve:

- **Sympathomimetics** for appetite suppression (anorexiant), indicated for short-term use
- **Lipase inhibitors** for blockage of dietary fat absorption, indicated for long term use



Please see nurse achieve lecture "Pharmacology" for additional drug information



Nursing Interventions:

- Patients must understand that drug therapy will not cure obesity- lifestyle modifications are essential to maintain weight loss
- The client should avoid over the counter diet aids
- The client should be made aware of relevant side effects and administration.

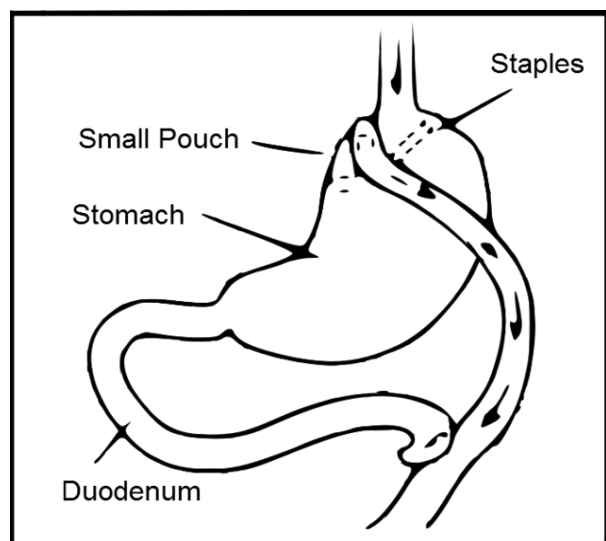
COPYRIGHT © 2025 NURSEACHIEVE, CORP.

15

OBESITY - SURGICAL INTERVENTIONS

Bariatric surgery is the surgical treatment for extremely obese clients (BMI > 40 kg/m²).

- **Restrictive surgeries** – reduces stomach size or restricts stomach volume (gastric banding, vertical sleeve gastrectomy)
- **Malabsorption procedures**- interferes with the absorption of food and nutrients (biliopancreatic diversion)
- **Combination** – both restricts and inhibits absorption (**gastric bypass surgery**)



COPYRIGHT © 2025 NURSEACHIEVE, CORP.

16

OBESITY - SURGICAL INTERVENTIONS

Bariatric Post-surgical Complications

- Anastomotic Leaks
 - **Manifestations:** Increasing back and abdominal pain, restlessness, tachycardia, oliguria
- Nutritional deficiencies and dehydration
- Dumping syndrome: stomach contents move into small intestine rapidly, and draw fluid in
 - **Manifestations:** nausea, bloating, abdominal pain, weakness, sweating, syncope

Discharge Teaching

- Diet and exercise regimen
- Nutritional supplements and hydration plan
- Signs and symptoms of infection and other complications
- Potential for psychological complications or body image issues (due to excess skin)

COPYRIGHT © 2025 NURSEACHIEVE, CORP.

17

METABOLIC SYNDROME - DEFINITION and DIAGNOSIS



Metabolic Syndrome: a group of metabolic risk factors that increase the risk of developing cardiovascular disease, stroke, and diabetes.

Fasting blood glucose: ≥ 100 mg/dL or on drug treatment for elevated glucose

Waist circumference: >40 in. or 102 cm (men) >35 in. or 89 cm (women)

HDL cholesterol: <0.9 mmol/L (<40 mg/dL) in men,
 <1.1 mmol/L (<50 mg/dL) in women

Triglycerides: >1.7 mmol/L (>150 mg/dL) or on associated drug treatment

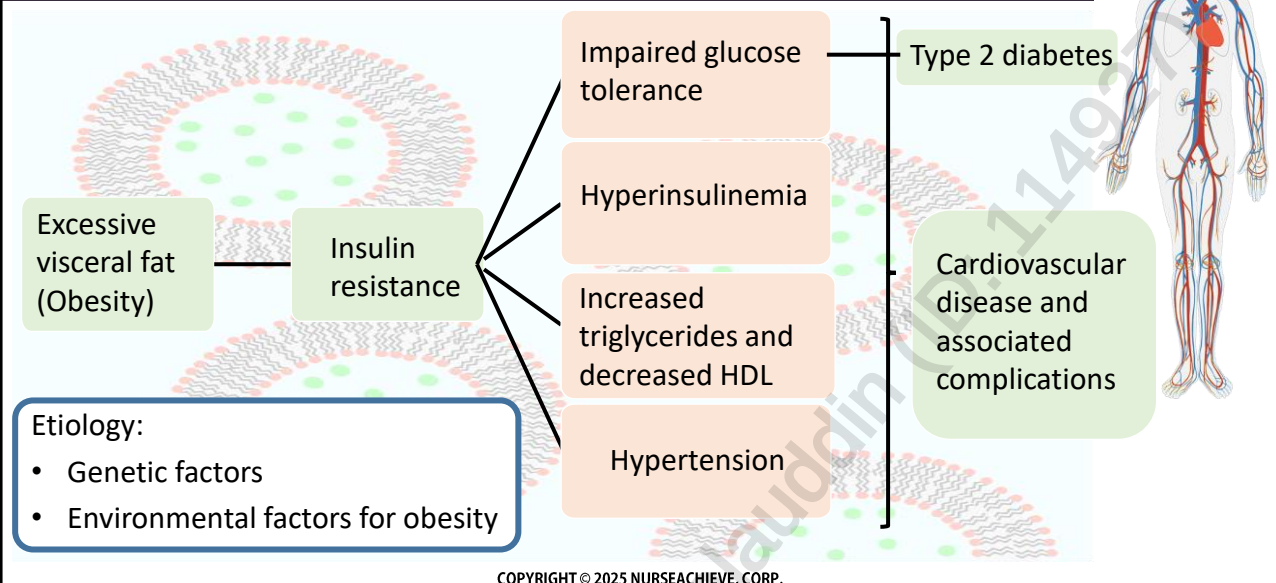
Blood pressure: SBP >130 mm Hg or DBP >85 mm Hg or on associated drug treatment

3 or more diagnostic factors listed above confirms **metabolic syndrome**

COPYRIGHT © 2025 NURSEACHIEVE, CORP.

18

METABOLIC SYNDROME - PATHOPHYSIOLOGY AND ETIOLOGY



19

METABOLIC SYNDROME - MANAGEMENT

- 1** Lifestyle modifications to reduce risk factors of cardiovascular disease: lower LDL cholesterol, stop smoking, lower BP, reduce glucose intake
 - 2** Long term lifestyle changes: diet, exercise, and weight reduction
 - 3** Cholesterol lowering and anti-hypertensive drugs may be indicated
- There is no specific management of metabolic syndrome; care most often prioritizes lifestyle changes for weight reduction and management

COPYRIGHT © 2025 NURSEACHIEVE, CORP.

20

ADULT MALNUTRITION - DEFINITIONS and ETIOLOGY

Malnutrition: an imbalance of essential nutrients (macro and micronutrients)

- **Undernutrition:** a state of inadequate or deficient nutrients as a result of inadequate diet or diseases that interfere with normal appetite

Primary malnutrition (starvation):

occurs when nutritional needs are not met due to inadequate intake

- poverty, substance abuse, decreased functional ability, lack of education, dementia, eating disorders

Secondary malnutrition (disease related):

Underlying diseases ↓ appetite or absorption of nutrients; tissue needs not met, leads to compromise of growth

- chronic conditions such as organ failure, cancer, and HIV
- acute conditions such as major infection and burns
- food-drug interactions

COPYRIGHT © 2025 NURSEACHIEVE, CORP.

21

ADULT MALNUTRITION - PATHOPHYSIOLOGY

1. Glycogen is used from carbohydrate stores in liver and muscle until depletion

2. Gluconeogenesis: skeletal protein is converted to glucose for energy

3. Fats are broken down into free fatty acids and ketones to supply brain nutrients

4. With prolonged starvation (after 5 days), body fat is primarily used, and muscle is conserved

- Liver function impaired, lack of plasma proteins results in fluid and electrolyte shifts and edema
- Malabsorption in GI mucosa
- Decreased myocardial contractility and CO
- Compromised respiratory function

5. Visceral proteins from organs and plasma must be used when fat stores are depleted

Organ failure and eventual death

COPYRIGHT © 2025 NURSEACHIEVE, CORP.

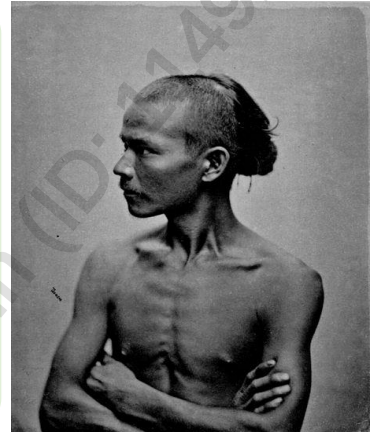
22

ADULT MALNUTRITION - MANIFESTATIONS

Specific clinical manifestations will vary regarding specific nutrient deficiencies

General undernutrition may manifest with the following symptoms depending on severity:

- **Neurological:** listlessness, confusion, irritability, drowsiness
- **Integumentary:** brittle and sparse hair, dry and inelastic skin with discoloration, cyanosis, peripheral edema
- **Respiratory/Cardiovascular:** decreased respiratory rate, decreased BP
- **Gastrointestinal:** decreased bowel sounds, ascites, distension, swollen oral mucosa, gums receding
- **Musculoskeletal:** muscle wasting, poor tone, prominent bony structures



COPYRIGHT © 2025 NURSEACHIEVE, CORP.

23

ADULT MALNUTRITION - GOALS OF CARE

- 1 Identify and treat adverse effects of malnutrition
- 2 Restore body mass, particularly muscle mass
- 3 Replace depleted nutrients and minerals

COPYRIGHT © 2025 NURSEACHIEVE, CORP.

24

ADULT MALNUTRITION - INTERVENTIONS

NUTRITION MANAGEMENT: careful reintroduction of fluids and nutrients is essential

Refeeding can precipitate electrolyte imbalances, heart failure, malabsorption, and diarrhea

1. Correction of fluid and electrolyte imbalances
2. Gradual re-introduction of protein and calories
3. Vitamin and mineral supplements
4. Fat and lactose are introduced last

Enteral and parenteral nutrition may be used to meet initial calorie and protein requirements

Nursing Alert!

Practical Nurses can not initiate TPN for a client or monitor a client on TPN. This task can only be completed by a Registered Nurse

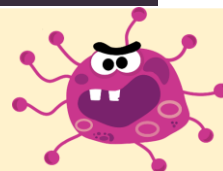
COPYRIGHT © 2025 NURSEACHIEVE, CORP.

25

ADULT MALNUTRITION - COMPLICATIONS & INTERVENTIONS

Infection: malnutrition affects several immune system components, including the skin, mucous membranes, and lymph tissue

- Monitor temperature and infection manifestations
- Maintain medical asepsis

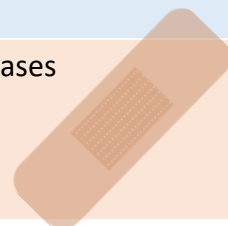


Deficient fluid volume: hyperosmolar nutritional solution use and difficulty swallowing may impair hydration

- Monitor oral mucous membranes and electrolyte balance
- Weigh daily and monitor intake and output
- Offer fluid frequently in small amounts

Impaired skin integrity: loss of subcutaneous tissue and muscle increases pressure ulcer risk

- Assess skin and turn/reposition every 2 hours
- Keep skin clean and dry, provide therapeutic surfaces



COPYRIGHT © 2025 NURSEACHIEVE, CORP.

26

SUMMARY

- BMI is a measure for determining relative body size and screening for nutritional disorders
- Obesity is a complex, multi-factorial disorder that results from hyperplasia and hypertrophy of adipocytes
- The complications of obesity are most commonly cardiovascular and respiratory, but can also include several other body systems
- Management of obesity primarily includes lifestyle modifications, and in extreme cases may include pharmacological and surgical therapy (bariatric surgery)
- Metabolic syndrome is a specific set of metabolic risk factors that pre-disposes a client to cardiovascular disease (management is similar to that of obesity)
- Adult malnutrition can be due to inadequate intake, chronic or acute illness. Starvation will eventually progress to death by organ dysfunction and failure
- Treatment for malnutrition requires careful refeeding and supplementation

COPYRIGHT © 2025 NURSEACHIEVE, CORP.

27

REFERENCES

1. Burchum, J. & Rosenthal, L. (2022). *Lehne's Pharmacology for Nursing Care*. Elsevier, Inc.
2. Harding, M., Kwong, J., Hagler, D., & Reinisch, C. (2023). *Lewis's Medical-Surgical Nursing: Assessment and Management of Clinical Problems*. Elsevier, Inc.
3. Ignatavicius, D.D., Rebar, C.R., Heimgartner, N.M. (2024). *Medical-Surgical Nursing: Concepts for Clinical Judgment and Collaborative Care*. Elsevier, Inc.
4. Jarvis, C. & Eckhardt, A. (2024). *Physical Examination & Health Assessment*. Elsevier, Inc.
5. Perry, A., Potter, P., Ostendorf, W., & Laplante, N. (2022). *Clinical Nursing Skills and Techniques*. Mosby.

COPYRIGHT © 2025 NURSEACHIEVE, CORP.

28